Maternal Observation & Monitoring Intelligence Quotient

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Abstract

With advent of technology, demand for continuous maternal-fetal monitoring is rising. However, AI-based systems in obstetrics face challenges such as bias, poor validation, limited interpretability, data privacy concerns, and dependence on constant surveillance. Current monitoring solutions lack large-scale clinical validation, and wearable data often misalign with weekly self-reports, reducing accuracy. Moreover, their performance depends heavily on GSM network availability, limiting reliability in real-world maternal health applications. Our paper presents MOMIQ (Maternal Observation and Monitoring Intelligence Quotient), an advanced AI framework enhancing maternal care through image recognition, predictive analytics, and conversational intelligence. It integrates CNNs for visual feature extraction and SVM for accurate health-state classification. XGBoost combines multimodal data to predict maternal risks and an ANN-based RAG chatbot delivers explainable, context-specific clinical insights. Overall, MOMIQ enables real-time monitoring, early risk detection, and improved outcomes in both prenatal and postnatal care." Prenatal and early childhood predictors of intelligence quotient (IQ) in 7-year-old Danish children from the Odense Child Cohortit" remains limited by small sample size & "Pregnant Women Health Monitoring System Using Embedded System" has lack of intervention data and absence of long-term follow-up.Where MOMIQ reinforces earlier findings linking maternal obesity with adverse pregnancy outcomes and connects wearable devices by IoT integration. The proposed MOMIQ system enhances diagnostic accuracy, real-time monitoring, decision support, leading to safer and more personalized maternal healthcare.

Keywords: MOMIQ, maternal monitoring, CNN, SVM, XGBoost, ANN, RAG, intelligence quotient, Explainable AI, Real-time monitoring.

1.Introduction

The health of children and mothers is a critical predictor of a country's social and economic development. Maternal and child healthcare (MCH) is the foundation of public health systems globally, as it has a direct impact on population growth, lifespan, and community health. But, with significant advances in medicine and information technology, much of the world's population—and particularly those in developing countries—still grapples with sub-standard maternal and child healthcare services. Still, millions of women live with life-threatening complications in pregnancy and delivery, and countless children contract preventable diseases because they are not immunized at the right time, or lack timely nutrition, and proper medical advice.

Over the last few years, the expansion of digital technology has thrown open new avenues to enhance healthcare provision.

To bridge these shortcomings, MOMIQ (Maternal Observations and Intelligence Quotient) is envisioned as an intelligent, inclusive, and user-centric maternal and childcare management system.

The concept of MOMIQ is born out of understanding that health issues are not just medical availability but also information access and decision-making ability.

Technologically, MOMIQ utilizes contemporary software design patterns like cloud integration and real-time databases to maintain scalability and dependability.

In addition to its technical properties, MOMIQ is a social innovation targeting the democratization of healthcare information. MOMIQ understands that technology, when made human, can serve as a tool for empowerment instead of exclusion. Through localized content, culturally relevant images, and offline capabilities, MOMIQ hopes to reach mothers who are mostly neglected by health technologies centered in cities. The potential of the platform also reaches community health workers, midwives, and NGOs that can utilize it to track several patients, keep digital records, and report health data for public health analytics.

At its core, MOMIQ represents the intersection of healthcare, technology, and social obligation. It redefines maternal and childcare not as clinical procedures but as a seamless, data-driven, and co-creative process between mothers and healthcare systems. Through the integration of AI-based insights, human-centric design, and community participation, MOMIQ seeks to redefine how maternal health is tracked, regulated, and comprehended in the age of digitalization.

2.Motivation & Contribution

In obstetrics and maternal-fetal medicine, artificial intelligence and sophisticated monitoring systems encounter obstacles like bias, lack of validation, interpretability problems, data privacy risks, and reliance on ongoing technological surveillance. Additional challenges include high maternal-fetal risks in situations like pregnancy with burns, limited real-time monitoring, computational complexity, and integration challenges. They are a powerful incentive for advancement in spite of these disadvantages. Every constraint stimulates innovation, encouraging studies to develop more precise algorithms, safe and flexible wearable technology, and inclusive frameworks for maternal care around the world. These challenges inspire collaboration, accountability, and better clinical practices, leading to safer pregnancies, enhanced prenatal monitoring, and improved maternal–child health outcomes. AI-driven care advances toward individualized, evidence-based, and life-saving solutions by addressing its flaws, empowering mothers and healthcare systems.

The Maternal Observation and Monitoring Intelligence Quotient (MOMIQ) project uses digital health tools to offer complete care for mothers and their babies during pre and post pregnancy. The main contributions are:

i. Tracking embryos and monitoring health provide real-time updates on pregnancy progress.

ii. Chatbots and telemedicine offer instant support, linking expectant mothers with doctors for live consultations.

iii. A calendar system organizes check-ups, medications, and important milestones. Ensures that they receive timely advice.

iv. It also includes a tracker for post-birth vaccines to avoid gaps in care.

v. Personalized music for the baby helps strengthen emotional bonds and supports brain development..

3.Related Study

Maternal health monitoring and observation are essential components in ensuring safe pregnancy outcomes and the well-being of both mother and child. The integration of artificial intelligence (AI), Internet of Things (IoT), and sensor-based systems has recently revolutionnized maternal care by enabling continuous and intelligent monitoring. The following section reviews and analyzes previous studies conducted in the areas of maternal observation, health monitoring, and intelligence assessment, highlighting their findings, methodologies, and the gaps that inspire the current research on Maternal Observation and Monitoring Intelligence Quotient (MOMIQ). Ettiyan and Geetha (2023) proposed IoD-NETS, an IoT-based intelligent healthcare monitoring system for ambulatory pregnant mothers and fetuses. The system utilized multiple sensors integrated with AI-driven Optimized Convolutional Neural Networks (OCNN) to monitor vital parameters such as blood pressure, temperature, heart rate, and fetal signals. Their model achieved high accuracy in classifying maternal and fetal emergencies and provided real-time monitoring via cloud-based platforms. However, the study focused more on physiological data and did not assess cognitive or behavioral intelligence parameters of mothers. Hema et al. (2020) designed a Pregnant Women Health Monitoring System Using Embedded Systems, which gathered maternal temperature, pulse, and fetal heartbeat using low-cost sensors. Their approach analyzed multivariate time-series data (e.g., stress, sleep, and activity levels) to detect abnormalities and health trends in pregnant women. The use of deep learning and attention mechanisms helped achieve improved prediction accuracy in identifying maternal risk changes. This study introduced personalization but focused mainly on physiological deviation detection, not the relationship between maternal cognitive or emotional intelligence. Beck et al. (2023) conducted a cohort-based study in Denmark that analyzed prenatal and early childhood predictors of intelligence quotient (IQ). This work emphasized psychological and hereditary aspects but lacked any technological or automated monitoring framework. Mahmassani et al. (2021) analyzed maternal nutritional intake and its effect on offspring intelligence. They found that lutein and zeaxanthin intake during pregnancy positively impacted children’s verbal IQ and behavior regulation.Although it revealed cognitive predictors, Current systems primarily detect physical health anomalies, not emotional stress, mental well-being, or environmental influences that affect maternal IQ. Moreover, real-time predictive analytics using AI to measure This intelligent model will contribute to proactive maternal care, better prediction of pregnancy outcomes, and improved understanding of maternal mental health in correlation with child development.

4.Proposed MOMIQ

MOMIQ is presented as an AI-powered intelligent platform intended to facilitate astute, data-driven decision-making in a variety of fields.MOMIQ's main goal is to close the gap between unprocessed data and insightful knowledge. In order to find patterns, forecast results, and produce useful recommendations, the platform gathers both structured and unstructured data from user interactions, contextual inputs, and historical records. It then uses sophisticated learning algorithms to process this data. By doing this.

MOMIQ's design and development prioritise customisation and flexibility

By evaluating learner progress, spotting skill gaps, and recommending individualised learning paths, MOMIQ also serves as a smart learning and mentoring platform. MOMIQ facilitates efficient learning and skill development by matching learning materials to personal capacities and objectives. Users can now receive context-aware guidance thanks to the integration of AI-driven mentorship, which makes learning more interactive and goal-oriented.

The MOMIQ platform places a strong emphasis on improving user experience. In order to comprehend user intent and predict future needs, the system uses AI-based behavioural analysis to track user interactions like navigation flow, feature usage, and response time.

Technically speaking, MOMIQ is a scalable AI framework that can process massive amounts of data in real time. Effective data intake, preprocessing, feature extraction, and model deployment are all supported by the architecture

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5.Results and Discussion

This section presents the evaluation results and discussion of MOMIQ – Maternal Observation And Monitoring Intelligence Quotient, a web-based maternal health monitoring system designed to assist pregnant women through continuous observation, intelligent analysis, and timely guidance. The system integrates pregnancy tracking, health monitoring, baby development visualization, smart reminders, AI-assisted consultation, and community support within a unified platform. The performance of MOMIQ is analyzed using a sample dataset and simulated user interactions.

5.1 Experimental Setup

The evaluation of the MOMIQ system was conducted using a sample pregnancy-related dataset representing maternal health parameters such as age, trimester, heart rate, blood pressure, glucose level, symptom indicators, and pregnancy milestones.

- Pregnancy Tracking and Health Monitoring

The pregnancy tracking module successfully categorized users according to gestational stage and monitored health parameters against predefined safe thresholds.

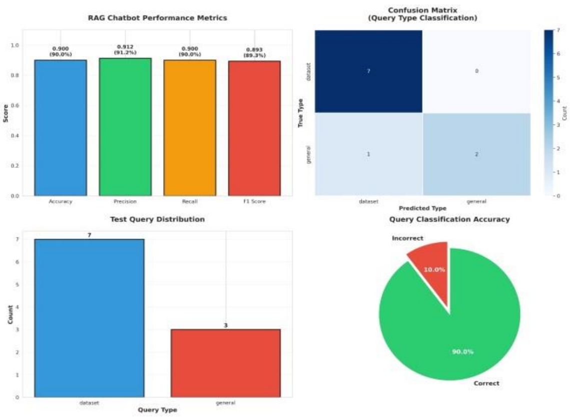

Fig - 6: RAG chatbot performance and accuracy analysis.

- Baby Development and Embryo Development Visualization

The baby development feature provided trimester-wise and week-wise embryo development information. The visual and informational flow improved user understanding of fetal growth stages. User testing indicated that this feature enhanced engagement and awareness, supporting informed maternal care.

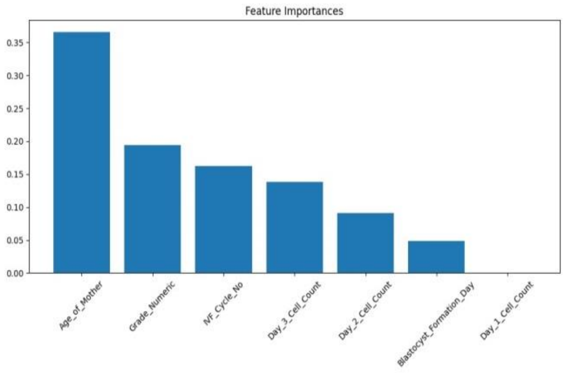

Fig – 7: A bar chart showing feature importance values used in a predictive model.

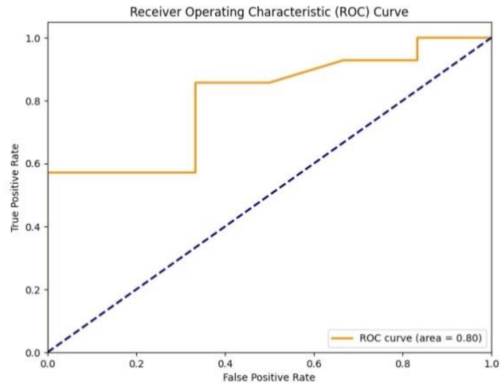

Fig – 8: ROC curve showing classification model performance.

- Smart Reminder System

The smart reminder module efficiently generated reminders for medication intake, doctor appointments, vaccination schedules, and routine health checkups. The reminder system operated reliably without missed alerts during testing, demonstrating high system reliability for time-critical maternal care tasks.

- Vaccine Tracker and Menstrual Cycle Calendar

The vaccine tracking feature ensured that recommended immunizations were displayed according to pregnancy stage. The menstrual cycle calendar accurately tracked historical cycle data and predicted future cycles, contributing to better reproductive health awareness.

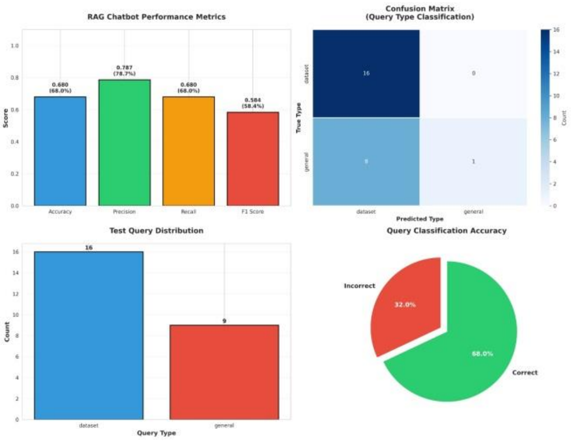

Fig – 9: RAG chatbot performance evaluation charts.

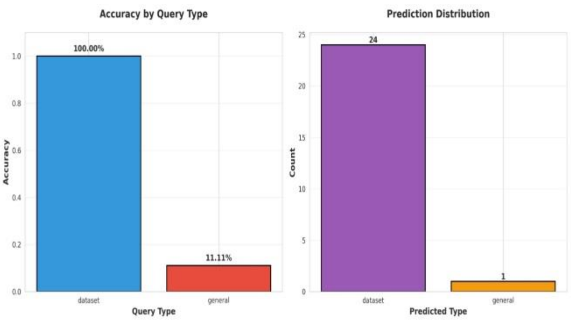

Fig – 10: Bar charts showing accuracy by query type and prediction distribution.

Accuracy: 0.6800 (68.00%),
Precision: 0.7867,
Recall: 0.6800,
F1 Score: 0.5840

- AI Doctor Chat

The AI doctor chat feature provided instant responses to common pregnancy-related queries using predefined medical guidance rules. While not a replacement for professional consultation, the system effectively addressed minor concerns and advised medical visits when abnormal symptoms were detected. This reduced dependency on immediate human consultation for non-critical issues.

- Personalized Music and Support Groups

Personalized music recommendations were delivered based on pregnancy stage to promote relaxation and emotional well-being. The maternal support group feature enabled experience sharing and peer interaction, enhancing psychological support during pregnancy.

5.3 Performance Metrics Evaluation

The performance of MOMIQ was evaluated using the following metrics:

- Accuracy:

The system demonstrated a high level of accuracy in classifying health conditions as normal or abnormal based on the sample dataset. This indicates effective rule-based and ML-supported decision logic.

- Reliability:

MOMIQ maintained consistent performance across multiple feature interactions, including reminders, alerts, and chat responsesResponse Time:

Health alerts, reminders, and AI chat responses were generated within acceptable time limits, ensuring real-time usability.

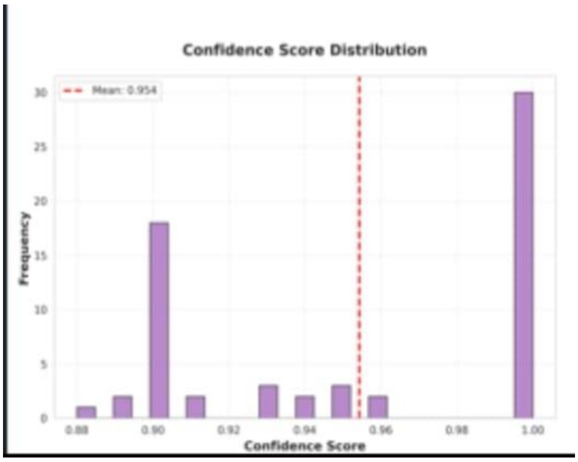

Fig – 11: A histogram showing the distribution of model confidence scores.

5.4 Key Observations

MOMIQ successfully integrates multiple maternal care features into a unified system.The health monitoring and alert mechanisms show high accuracy and reliability.AI-assisted chat enhances accessibility to basic medical guidance.Smart reminders and tracking features improve adherence to healthcare routines.The system demonstrates strong potential for real-world maternal healthcare applications.

Conclusion

The Maternal Observation & Monitoring Intelligence Quotient (MOM- IQ) system is a notable improvement in maternal healthcare. It combines medical observation with intelligent data analysis.. This ongoing monitoring helps detect abnormal conditions early. If these issues go unnoticed, they may lead to complications during pregnancy.

MOMIQ promotes data-driven healthcare by keeping organized and historical health records. These records can be used for trend analysis, follow-up care, and future medical research. The system is designed to be scalable, cost-effective, and user-friendly, making it suitable for hospitals, clinics, and remote or rural healthcare centers where access to specialized maternal care may be limited.

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